

ORIGINAL RESEARCH

A population-based recalibration method for updating survival neural networks models for cardiovascular risk prediction in United Kingdom and China

Xiaofei Liu^a, Jun Zhang^b, Larry Liu^{c,d}, Xun Tang^{a,e,f,*}, Pei Gao^{a,e,f,*}

^aDepartment of Epidemiology and Biostatistics, School of Public Health, Peking University, Beijing, China

^bMSD R&D (China) Co., Ltd., Beijing, China

^cMerck & Co., Inc., Rahway, NJ, USA

^dWeill Cornell Medical College, New York City, NY, USA

^eCenter for Real-World Evidence Evaluation, Peking University Clinical Research Institute, Beijing, China

^fKey Laboratory of Epidemiology of Major Disease (Peking University), Ministry of Education, Beijing, China

Accepted 25 June 2025; Published online 30 June 2025

Abstract

Objectives: Machine learning algorithms, particularly survival neural networks (SNNs), promise to improve cardiovascular disease (CVD) risk prediction. However, the necessity and approach for recalibrating SNN models across diverse populations remain unclear. We aimed to propose a population-based recalibration method and validate it using two large cohort studies.

Study Design and Setting: A total of 347,206 individuals aged 40–74 years without prior CVD from UK Biobank (UKB) were for model training and internal validation and 177,756 individuals from a Chinese cohort study (CHinese Electronic health Records Research in Yinzhou [CHERRY]) were for external validation. Three types of SNN models (DeepSurv, age-specific DeepSurv, and DeepHit) were developed for the 10-year CVD risk prediction and compared to Cox models. These models were recalibrated using the proposed method to adjust for differences in disease incidence between populations based on population-level summarized data, and compared with a traditional method that used individual-level data.

Results: All SNN models demonstrated robust discrimination in both UKB and CHERRY validation sets (C-indices > 0.720), but underpredicted risk for CHERRY populations by 60%. The population-based recalibration method largely corrected the initial risk underestimation, yielding observed-to-expected (O:E) ratios of 1.080, 1.115, and 1.153 for recalibrated DeepSurv, age-specific DeepSurv, and DeepHit, achieving comparable calibration to individual-based recalibration method (O:E ratios: 1.040, 1.054 for DeepSurv and age-specific DeepSurv). The well-calibrated age-specific DeepSurv and Cox models identified high-risk groups with distinct characteristics, with 79% overlap for women and 63% for men.

Conclusion: The proposed method effectively adjusts predictions for survival neural network models using population-level summarized data without modifying the original network, making recalibration essential for applying machine learning models to different populations. The method highlights the clinical potential of SNN models for broader application across diverse regions. © 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Keywords: Machine learning; Survival neural network; Model updating; Prediction model; Recalibration; Cardiovascular disease

Funding: This study was supported by the Noncommunicable Chronic Diseases-National Science and Technology Major Project of China (2023ZD0503500), the Beijing Natural Science Foundation (grant number: IS24047), the National Natural Science Foundation of China (grant number: 82373662) and Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

Ethics statement: The UK Biobank study was approved by the Northwest Multi-center Research Ethics Committee as a Research Tissue Bank (RTB) approval (REC reference: 21/NW/0157) and all participants provided

written informed consent. This study was conducted under RTB approval with project ID 99040. The CHERRY study was approved by the Peking University Institutional Review Board (IRB00001052-16011).

* Corresponding author. Department of Epidemiology and Biostatistics, Peking University Health Science Center, 38 Xueyuan Road, Beijing 100191, China.

E-mail addresses: tangxun@bjmu.edu.cn (X. Tang); peigao@bjmu.edu.cn (P. Gao).

Plain Language Summary

Machine learning, particularly neural networks for survival analysis, shows great potential in disease risk prediction but typically requires adjustments, or "recalibration" for diverse populations. We proposed a recalibration method using population-level summarized data, rather than individual data, which is often hard to obtain. We derived several survival neural network models for 10-year cardiovascular disease risk prediction in the UK Biobank and validated them in the CHinese Electronic health Records Research in Yinzhou (CHERRY) cohort. Although the models ranked individual risk effectively, they underpredicted actual risk in the Chinese population. The proposed method successfully corrected this underprediction, aligning model outputs with observed risk. This method works for different types of survival neural network models and preserves the original model structure. This advancement enhances the reliability and generalizability of survival neural network models across regions, underscoring the need for model evaluation and recalibration before applying them to new populations. It offers a practical solution for improving their accuracy without needing detailed individual data, thus broadening the clinical application of survival neural network models in disease prevention.

1. Introduction

Cardiovascular disease (CVD) risk assessment is critical for clinicians to make informed decisions regarding primary prevention treatments based on an individual's risk profile [1]. Most guideline-recommended CVD risk prediction models, such as the Framingham Risk Score (FRS) [2] and the World Health Organization (WHO) risk charts [3], have been developed using traditional statistical regression methods. Recently, machine learning (ML) algorithms, particularly neural networks, are increasingly being used for disease risk prediction due to their ability to analyze complex patterns in large clinical datasets [4]. The integration of neural networks into survival analysis has given rise to survival neural networks (SNNs), which are well-suited to handle censored medical data. The SNNs could be categorized based on the underlying statistical technique that forms their foundation, such as the Cox-based (DeepSurv) [5], discrete-time (DeepHit) [6,7], and other types [8]. An increasing number of studies have used SNNs to develop risk prediction models that incorporate various clinical features, demonstrating promising performance in predicting CVD events [9–14].

However, many of these studies, which often use data from a single or geographically similar cohort [12–14], often lack rigorous external validation. Evidence shows that when traditional regression-based models are applied to different populations with distinct risk factors, interpopulation heterogeneity can lead to systematically inaccurate risk assessments, potentially causing overtreatment or undertreatment in risk-based interventions such as statin therapy [15,16]. The challenge of maintaining prediction accuracy across diverse populations is also prevalent in SNN models. Therefore, proper model validation is essential to ensure precise prediction and enable successful clinical implementation [17]. Moreover, recalibration may offer a potential solution to transitioning models between populations, rather than discarding the model or developing a new one [18]. However, most existing recalibration methods focus on traditional statistical models and are primarily applicable to Cox

structured models, leaving a gap in research on recalibrating SNN models. Consequently, it remains unclear whether SNN models require recalibration across populations and, if so, how to recalibrate them effectively.

In this study, we aim to derive and validate SNN models for 10-year CVD risk prediction using two large-scale population-based studies from the United Kingdom and China. Our objectives are to compare the performance of the SNN model under different training strategies with the traditional Cox proportional hazards model as the reference and to investigate recalibration methods for different types of survival ML models when applied to external populations.

2. Methods

2.1. Study population

We developed SNN models using data from the UK Biobank (UKB) cohort study [19], and validated them in the CHinese Electronic health Records Research in Yinzhou (CHERRY) study [20] (Fig 1). The UKB study enrolled 502,413 participants aged 40–74 years from 22 assessment centers across England, Wales, and Scotland for the initial assessment visit between 2010 and 2016. For model development, we excluded participants who (i) had a prior diagnosis of CVD at enrollment or (ii) lacked measurements of key predictors, including age, sex, smoking status, systolic blood pressure (SBP), high-density lipoprotein cholesterol (HDL-C), total cholesterol (TC), and a history of diabetes, as in the FRS [2]. We defined the baseline as the date of initial assessment and chose the baseline assessment for each predictor.

The CHERRY study is an electronic health record (EHR)-based cohort derived from the integrated health information system in the Yinzhou district of Zhejiang Province, a developed region in southeastern China [20]. The CHERRY study enrolled 1,053,565 Chinese adults in 2009 by linking multiple data sources, including population census, registered health insurance, health checks, inpatient and outpatient electronic medical records, disease

What is new?**Key findings**

- Survival neural network (SNN) models demonstrated consistent discrimination but varied calibration for 10-year cardiovascular risk prediction in two population-based cohort studies from the United Kingdom and China.
- A new recalibration method, based on differences in disease incidence between populations, effectively corrected model misestimation.

What this adds to what is known?

- Machine learning algorithms hold great potential in disease prediction but remain underutilized in clinical practice due to limited external validation and challenges in model updates for generalizability. This study addressed this gap by proposing a new recalibration method for different types of SNN models using population-level summarized data without modifying the original network.

What is the implication and what should change now?

- SNN models need to be evaluated with caution before applying them to new populations.
- The population-based recalibration method could enhance the generalizability of survival machine learning models and expand their potential for broader clinical use in disease prevention.

surveillance and management data, and death certificates. We extracted participants for external validation using the same criterion as for the training set. Referring to EHR-based studies, such as the QRISK3 equations in the UK QResearch database [21], we defined individual-specific baseline as (i) the date of registration in the health information system, (ii) the date of the participant's 40th birthday, (iii) the date of the first recorded serum lipid measurement, or (iv) the study start date of January 1, 2010, whichever was the latest. For each risk factor with multiple measurements, values closest to the baseline for each participant were chosen for analysis.

2.2. Outcome definition

The primary outcome in our study is the first-ever onset of CVD, including nonfatal myocardial infarction, fatal coronary heart disease, and fatal or nonfatal cerebrovascular disease. Follow-up was censored at the date of first CVD diagnosis, when lost to follow-up due to any cause of death, or at the date of data updating (May 31, 2021 for UKB and

May 31, 2020 for CHERRY), whichever occurred first. A detailed comparison of the measurement method for predictors and the outcome for two cohorts are provided in [Supplementary Table 1](#).

2.3. Risk prediction models

We developed SNN models using two types of algorithms. The DeepSurv is a fully connected single-output neural network that computes the individual hazard functions and combines with Cox regression to calculate the 10-year CVD risk [5]; the DeepHit is a fully connected multioutput neural network that estimates the distribution of survival probabilities over discrete time intervals [6]. Five risk prediction models with the same seven predictors as the FRS were developed: traditional DeepSurv, age-specific DeepSurv, DeepHit, simple Cox, and Cox with age interactions.

The traditional DeepSurv and DeepHit models were developed in the entire training set, and the age-specific DeepSurv model was developed separately in each age group (<50 years, 50–65 years, and ≥65 years). The simple Cox model, incorporating core traditional risk factors, was a best-fit version of the FRS with UKB study-specific coefficients, which were sex-specific. It served as a clinically interpretable benchmark to facilitate the evaluation of the SNN models and the proposed recalibration method. In the Cox with age interactions model, we included the quadratic term of age and the interaction term between age and the following predictors: SBP, TC, HDL-C, smoking status, and diabetes history, compared to the simple Cox. The UKB cohort was randomly divided into a training set (80%) and an internal validation set (20%), and the training set was further divided into 80% for training and 20% for hyperparameter optimization. Using the multiobjective tree-structured Parzen estimator algorithm, we searched for optimal hyperparameters listed in [Supplementary Table 2](#) to maximize Harrell's C index and minimize the expected-to-observed ratios. This involved 100 experiments, and the optimal values of hyperparameters are in [Supplementary Table 3](#).

2.4. Model recalibration

Two methods for model updating were recalibration using individual data and recalibration using summarized data. The former is the common method for the traditional Cox model and is only suitable for DeepSurv with a similar semiparametric Cox structure. We adjusted the baseline survival function separately for sex, of the original Cox and DeepSurv models, estimated by the Breslow estimator using each individual's survival information from the CHERRY ([Supplementary Methods](#)). The latter recalibration method is the adjustment of the predicted risk with re-scaling factors, which has been adopted by some global CVD models such as Systematic COronary Risk Evaluation

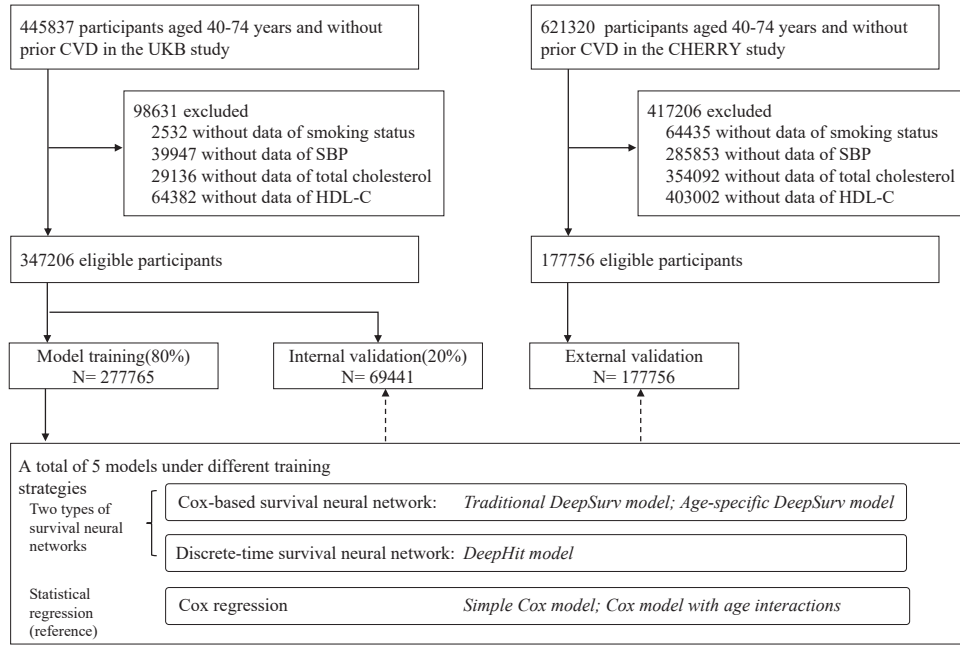


Figure 1. Flow chart of the study population.

2 (SCORE2) and WHO risk charts. We proposed a simplified version based on the incidence rate of the new population with the assumption of homogeneity of the relative effect of the risk factors, and this method is adaptable to

the different types of SNN models (Fig 2). Key steps for recalibration using summarized data include.

- (i) Estimation of the 10-year risk for the new population: Depending on the form of the aggregated data, there

General progress

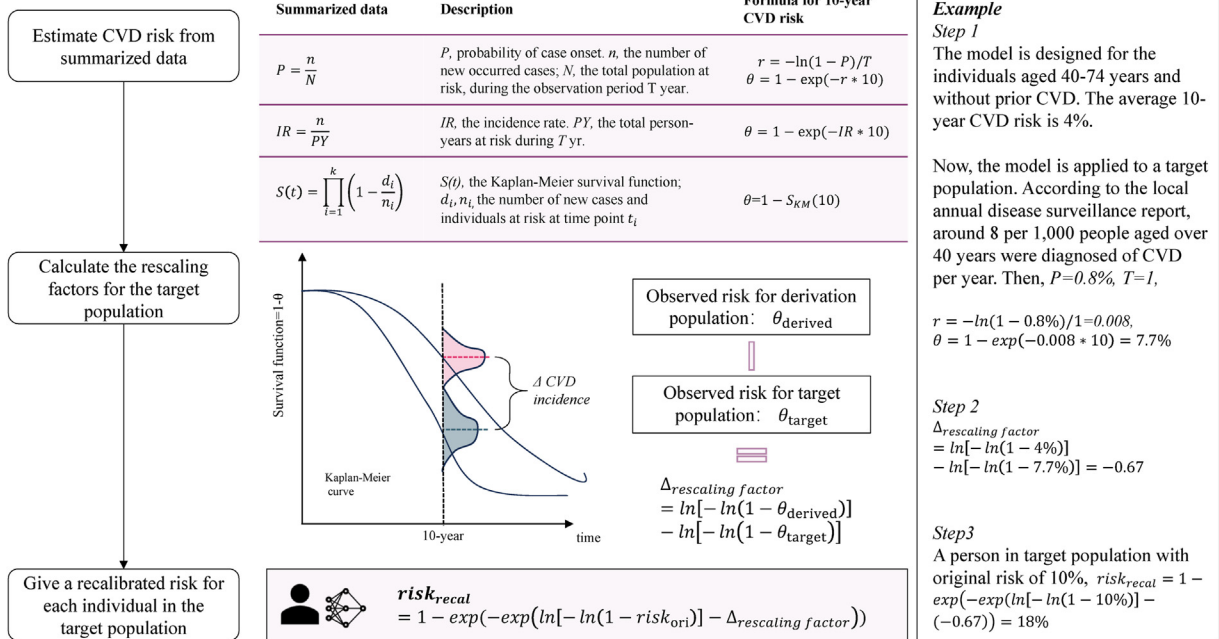


Figure 2. Process flow for the proposed recalibration method based on population-level summarized data. The population-based recalibration methods for survival neural network models can be applied in three key steps. An example of the recalibrated risk calculation is provided for illustration.

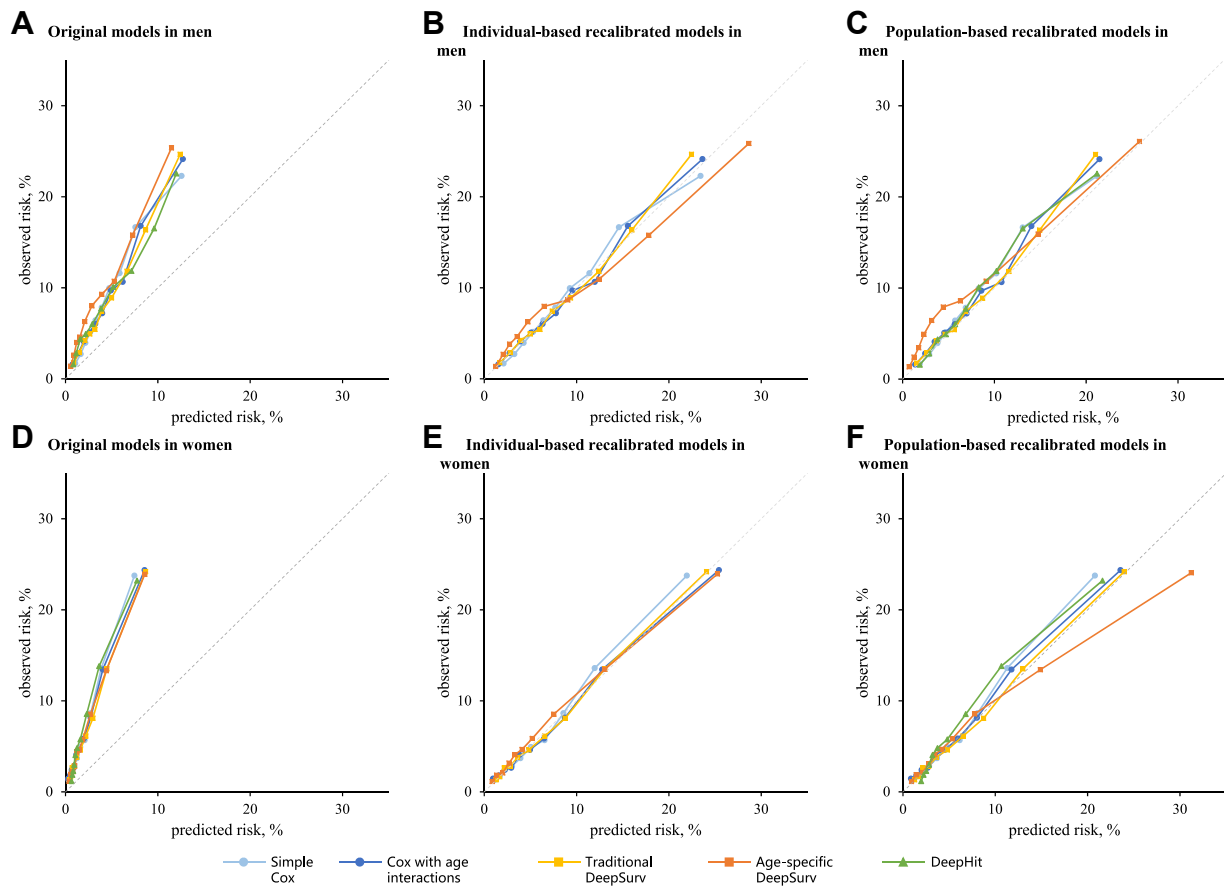


Figure 3. Calibration plot by deciles of predicted risk for 10-year risk in the CHERRY external validation. All original models (A, D), as well as the recalibrated models using the individual-based recalibration method (B, E) and the population-based recalibration method (C, F), were evaluated for calibration separately for men and women.

are several ways of converting rates into risks. For instance, the proportion of new-onset CVD for certain durations of time can first be converted to a 1-year incidence rate, or the incidence rate per person-year can be directly estimated, and then these rates can be projected to specific time periods (eg, 10-year risks), assuming that the annual incidence rate remains stable over time. In addition, given a stable cohort with records of events, specific time-period risks can also be estimated directly using the Kaplan–Meier estimator;

- (ii) Calculating the rescaling factors: This is performed by calculating the difference in transformed 10-year risks between the new population and the population in which the model was initially developed. For instance, in this study, we estimated the average observed 10-year risks for the entire UKB cohort (including training and internal validation datasets) and the entire CHERRY cohort (external validation dataset). The clog-log transformation was applied to both these values, and the difference was used as the rescaling factor. Depending on the available

information, rescaling factors may be specific to either sex or age. In this study, the Cox, traditional DeepSurv, and DeepHit models were recalibrated separately for sex using the new proposed population-based method. To simulate the application of this method with varying levels of accessible data, we further calculated rescaling factors by age group, thereby recalibrating the age-specific DeepSurv model ([Supplementary Table 4](#));

- (iii) Rescaling the predicted risk for each individual using the sex- or age-specific rescaling factor.

To explore the performance of the proposed recalibration method under three different theoretical miscalibration scenarios, we conducted a simulation study. Details of the simulation design and results are provided in the [Supplementary Methods](#).

2.5. Model evaluation

Based on the performance of the original SNN models in the UKB, the estimated minimum sample size required for external validation was 29,000 calculated using a

simulation-based approach [22] (details in [Supplementary Methods](#)), indicating that the CHERRY cohort is sufficient for the external validation. Discrimination was evaluated by the Harrell's C index and calibration was evaluated using ratios of observed-to-expected events (O:E ratio). We plotted the predicted risk against the observed risk for each decile of the predicted risk or each 5-year age group, as well as the smoothed calibration curves using a Cox model with restricted cubic splines [23]. Observed risk was calculated using the Kaplan–Meier method. To better evaluate the reclassification capacity, we calculated two types of categorical net reclassification improvement (NRI) using 10% as the threshold for 10-year CVD risk, a cut-off commonly used in international guidelines (eg., National Institute for Health and Care Excellence, United States Preventive Services Task Force, Chinese guidelines) to define high-risk individuals for intervention. One was for participants with no censoring within the 10-year follow-up [24]. The other, designed for survival data, evaluated correct classification for all participants, estimating expected events in censored individuals with Kaplan–Meier estimates [25]. The simple Cox model served as a standard, clinically interpretable benchmark for evaluating the performance gains of the SNN models. This study is reported following the transparent reporting of a multivariable prediction model for

individual prognosis or diagnosis (TRIPOD + AI) statement [26]. All analyses were conducted using Python (Python Software Foundation, version 3.7) and STATA (StataCorp LLC, version 14). A two-sided *P* values of less than 0.05 were considered statistically significant for all comparisons.

3. Results

3.1. Population characteristics

Of the original 502,413 individuals in the UKB, 347,206 were eligible for analysis. The mean age was 58 years and 45.3% were men ([Table 1](#)). During a median follow-up of 12.3 years, 17,104 CVD events were recorded. The 10-year Kaplan–Meier CVD incidence was 3.66% (95% CI: 3.60%–3.72%). There was no significant difference between training and internal validation datasets ([Supplementary Table 5](#)). In the CHERRY cohort, 177,756 individuals were included for external validation, with a mean age of 55 years and 46.4% were men. Compared to the UKB, individuals in the CHERRY had a higher proportion of diabetes, but lower SBP and TC. The duration of follow-up was shorter in the CHERRY, but the 10-year CVD incidence was significantly higher than UKB. The hazard ratios of the traditional CVD

Table 1. Baseline characteristics for the UK Biobank and CHERRY cohorts

Variables	UK Biobank (<i>N</i> = 347,206)	CHERRY (<i>N</i> = 177,756)
Sex: men, <i>n</i> (%)	157,157 (45.3)	82,535 (46.4)
Age at baseline, years	58.0 ± 9.6	55.0 ± 10.4
Age groups, <i>n</i> (%)		
< 50 years	82,037 (23.6)	54,893 (30.9)
50–65 years	197,623 (56.9)	95,460 (53.7)
≥ 65 years	67,546 (19.5)	27,403 (15.4)
Systolic blood pressure, mmHg	136.0 (124.5–149.5)	130.0 (120.0–140.0)
Total cholesterol, mmol/L	5.70 (5.00–6.50)	4.89 (4.29–5.53)
High-density lipoprotein cholesterol, mmol/L	1.40 (1.20–1.70)	1.27 (1.08–1.50)
Smoking status, <i>n</i> (%)		
Ever/current smoker	153,154 (44.1)	33,952 (19.1)
Non-smoker	194,052 (55.9)	143,804 (80.9)
Diabetes, <i>n</i> (%)	14,709 (4.2)	13,755 (7.7)
Incident cardiovascular outcomes		
Follow-up, years	12.3 (11.5–13.0)	7.8 (5.5–9.6)
Number of CVD cases, <i>n</i> (%)	17,104 (4.93)	9714 (5.46)
Incidence of CVD, per 10,000 person-year (95% CI)	41.09 (40.48–41.71)	74.91 (73.43–76.42)
10-year Kaplan–Meier CVD incidence, % (95% CI)	3.66 (3.60–3.72)	8.03 (7.86–8.21)

Data are presented for categorical variables with *n* (%) and continuous variables with mean ± SD or median (25th–75th).

Table 2. Model discrimination in the UK Biobank and CHERRY validation sets

Models	Harrell's C index (95% CI)	vs Simple Cox model	
		Difference of C index	P value
UKB internal validation			
Simple Cox	0.7313 (0.7235,0.7392)	-	-
Cox with age interactions	0.7327 (0.7249,0.7405)	0.0014 (0.0001,0.0027)	0.035
Traditional DeepSurv	0.7338 (0.7260,0.7416)	0.0025 (0.0008,0.0042)	0.004
Age-specific DeepSurv	0.7332 (0.7253,0.7410)	0.0019 (0.0001,0.0036)	0.040
DeepHit	0.7319 (0.7240,0.7397)	0.0006 (−0.0004,0.0016)	0.274
CHERRY external validation			
Original models			
Simple Cox	0.7288 (0.7239,0.7338)	-	-
Cox with age interactions	0.7332 (0.7282,0.7382)	0.0044 (0.0036,0.0052)	<0.001
Traditional DeepSurv	0.7358 (0.7308,0.7407)	0.0070 (0.0059,0.0080)	<0.001
Age-specific DeepSurv	0.7409 (0.7359, 0.7458)	0.0120 (0.0104,0.0136)	<0.001
DeepHit	0.7275 (0.7225,0.7324)	−0.0014 (−0.0020,−0.0008)	<0.001
Individual-based recalibrated models ^a			
Simple Cox	0.7364 (0.7315,0.7413)	-	-
Cox with age interactions	0.7395 (0.7345,0.7444)	0.0030 (0.0022,0.0039)	<0.001
Traditional DeepSurv	0.7408 (0.7358,0.7457)	0.0044 (0.0033,0.0055)	<0.001
Age-specific DeepSurv	0.7433 (0.7384,0.7483)	0.0069 (0.0056,0.0082)	<0.001
Population-based recalibrated models			
Simple Cox	0.7362 (0.7313,0.7412)	-	-
Cox with age interactions	0.7394 (0.7344,0.7443)	0.0032 (0.0023,0.0040)	<0.001
Traditional DeepSurv	0.7404 (0.7354,0.7453)	0.0042 (0.0031,0.0052)	<0.001
Age-specific DeepSurv	0.7388 (0.7338,0.7439)	0.0026 (0.0008,0.0044)	0.004
DeepHit	0.7340 (0.7290,0.7390)	−0.0022 (−0.0032,−0.0013)	<0.001

The simple Cox model was derived by sex based on age, smoking status, history of diabetes, SBP, TC, and HDL-C. The Cox model with age interactions was derived by sex based on age, smoking status, history of diabetes, SBP, TC, HDL-C, and their interactions with age. Traditional DeepSurv model derived in the entire training set. Age-specific DeepSurv model was derived in each age group (<50, 50–65, ≥65 years).

^a Individual-based recalibrated method is not suitable for the DeepHit.

risk factors were broadly comparable across both populations, with similar directions and effect sizes. However, the effect of age differed more markedly than that of any other factors (Supplementary Figure 1).

3.2. Discrimination

As shown in Table 2, The C-indices for the original Cox and SNN models were above 0.72 in both the UKB or CHERRY validation sets. For original models, the C-index of the simple Cox model in the CHERRY set was 0.7288 (95% CI: 0.7239–0.7338). By adding age interactions, the model achieved better discrimination with a C-index of 0.7332 (95% CI: 0.7282–0.7382), which was closer to the traditional DeepSurv. Compared to the simple Cox model, the traditional DeepSurv and age-specific DeepSurv model had higher C indices (difference in C index: 0.0070 and 0.0120, $P < .001$), and DeepHit had a lower C index (difference in C index: −0.0014, $P < .001$). As recalibration was applied stratified by sex, model discrimination within

men and women remained unchanged (Supplementary Table 6). However, minor fluctuations were observed in the overall C-index for the entire population after recalibration, likely reflecting the differential adjustments applied across sex strata. After population-based recalibration, the recalibrated age-specific DeepSurv showed better discrimination than the simple Cox in men, those aged over 65 years, diabetics, and ever/current smokers, with a significant difference in C-indices of 0.018, 0.0056, 0.0051, and 0.0106 (Supplementary Table 7).

3.3. Calibration

The original Cox and SNN models were well-calibrated in the UKB validation set for men and women (Supplementary Figure 2). However, they all significantly underpredicted the risk in the CHERRY validation set by about 60% on average (Supplementary Table 8). The underprediction was consistent across all risk deciles and was more pronounced in women compared to men (Fig 3). This

miscalibration pattern corresponded to the first simulated miscalibration scenario ([Supplementary Methods](#)), where the observed and predicted risks showed an approximate linear relationship with consistent miscalibration. Consistent with the simulation findings, the proposed population-based recalibrated method significantly improved calibration for the DeepSurv and DeepHit, with an average underprediction of about 10% in men (O:E ratios of 1.109 and 1.121, respectively) ([Fig 3](#), [Supplementary Figure 3](#)). The recalibrated age-specific DeepSurv model demonstrated the best calibration, exhibiting only 2.4% overprediction in women (O:E ratio: 0.977). When recalibrated using the traditional individual-based method, the Cox and DeepSurv models could accurately estimate the absolute risk, yielding the O:E ratios of 1.014 in men and 1.107 in women for the age-specific DeepSurv model. Notably, the predicted risk distributions after recalibration by the two methods were quite similar for both cases and non-cases ([Supplementary Figure 4](#)). The age-specific DeepSurv model showed better calibration in the top two groups (age over 65 years) in men than the other models ([Supplementary Figure 5](#)).

3.4. Reclassification and identification of high-risk individuals

We calculated the reclassification of the SNN models over the simple Cox model in the CHERRY cohorts. The NRIs calculated among noncensored individuals for

traditional DeepSurv over the simple Cox was 3.11% (95% CI: 2.48%, 3.73%), showing better upward reclassification for CVD cases for both sexes. In contrast, the NRIs for the age-specific DeepSurv over the simple Cox was 1.38% (95% CI: 0.62%, 2.14%), demonstrating improved reclassification for non-cases in men and for cases in women ([Supplementary Table 9](#)). Absolute reclassification counts are provided in the [Supplementary Table 10](#). The NRIs accounted for censoring showed similar results for comparison of paired models ([Supplementary Table 11–12](#)).

The shape of the 10-year predicted risk distribution was similar for each model in both the UKB and CHERRY, with the SNN models exhibiting sharper distributions than the Cox model, particularly for CVD cases ([Supplementary Figure 6](#)). When comparing the age-specific DeepSurv to the simple Cox model, the absolute predicted risks were consistent in low-risk groups (below 10%). For high-risk individuals (over 10%), 79% of women and 63% of men were identified by both models. The simple Cox model identified an additional 1209 women and 6010 men, whereas age-specific DeepSurv identified an additional 2938 women and 3101 men ([Fig 4](#)). In contrast, the Cox model with age interactions showed a higher overlap with age-specific DeepSurv and identified more individuals under 50 years (10.2% vs 2.6%) and with diabetes (20.5% vs 14.7%) than the simple Cox. Overall, the two Cox models tended to identify more individuals with elevated

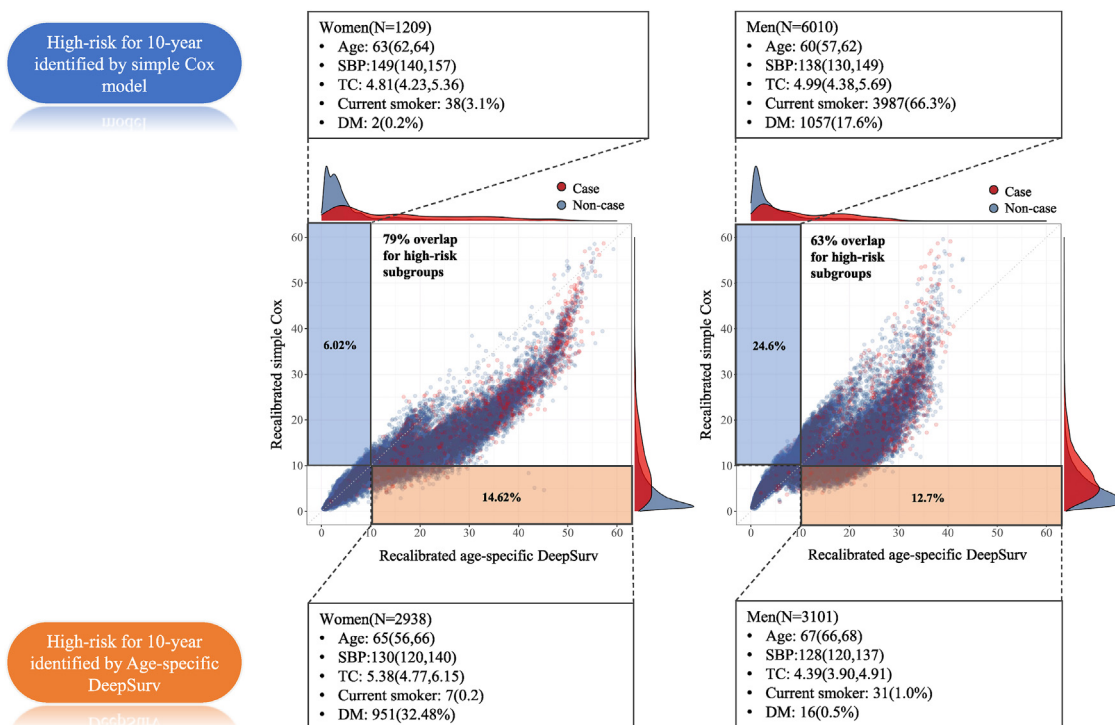


Figure 4. Reclassification of the simple Cox model vs the age-specific DeepSurv model in the CHERRY external validation. The scatter plot showed the predicted risk distributions by each model. Using a 10% cutoff, the blue area represented individuals identified as high risk solely by the simple Cox model, whereas the yellow area represented those identified as high risk only by the age-specific DeepSurv model. The characteristics of the high-risk populations were compared separately for men and women. DM, diabetes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

levels of SBP and current smokers, whereas the age-specific DeepSurv model identified more individuals aged over 65 years and more women with diabetes ([Supplementary Table 13–14](#)).

4. Discussion

In this study, different types of SNN models were developed in the UK cohort to predict 10-year CVD risk, although these models significantly underpredicted risk in the Chinese cohort for external validation. After recalibration using our proposed method, which relied on summarized CVD incidence data, the SNNs accurately predicted absolute risk, comparable to models recalibrated with individual-level data. These findings emphasize the importance of external validation and recalibration before clinical implementation. A recent review showed most ML models for CVD lack external validation, often being tested on datasets from the same country used for model development [9]. Even if ML models perform well internally, their performance may vary across different settings [23,27], such as in different geographic regions, in populations with different characteristics, or in different periods with the same region where risk factor distributions, population demographics, and health care practices may temporally change. Therefore, external and prospective validation is essential to ensure the continued accuracy and appropriateness of risk prediction models in clinical decision-making, such as guiding statin therapy for CVD prevention.

Calibration is a critical yet often overlooked metric in ML model evaluation, especially in risk-based treatment settings where accurate absolute risk estimates are essential [17]. Standard ML algorithms, which often assume censored individuals are event-free, tend to underestimate long-term risk. [28]. To address this, various ML methods tailored for survival data have emerged in recent years. These include tree-based models, such as survival trees and random survival forests [29], as well as multitask learning models for survival analysis. Among these approaches, SNNs are one of the most frequently used methods for disease risk prediction, with notable applications in cancer [30] and CVD [9]. In this study, we implemented two types of SNNs in this study to process the censoring information, each with its distinct merits. The SNNs ranked risks precisely but failed to account for baseline risk differences between the United Kingdom and China, likely due to ethnic, environmental, or other factors. As a result, they acted as an “uncalibrated classifier”.

Although some degree of miscalibration is often observed during external validation, its impact on clinical use hinges on its magnitude, pattern, and the context of clinical decision-making. The miscalibration occurring far away from specific thresholds might not alter clinical decisions and could still be useable. However, models providing absolute risk estimates used to guide interventions (eg, initiating

statin therapy based on a 10-year CVD risk threshold) require good calibration. Significant miscalibration near such clinical thresholds can lead to harmful decisions; underestimation may result in the undertreatment of individuals who could benefit, whereas overestimation may lead to unnecessary interventions, costs, and potential side effects. Updates are essential for poorly calibrated models. Retraining or revising models for optimal performance can be data-intensive and computationally demanding, whereas recalibration is a recommended approach for traditional regression models [31]; however, there is limited research on recalibrating survival ML algorithms [32]. Cox model recalibration typically adjusts baseline risk and predictor hazard ratios based on individualized data, a method applicable only to Cox structured SNNs like DeepSurv. Our population-based method builds on the principle of using aggregate data for recalibration, similar to methods used for the SCORE2 and WHO CVD risk charts. These models typically use region-specific incidence rates and risk factor means, often stratified by age and sex, within a regression framework to derive scaling parameters. Our proposed method offers a potentially simpler alternative, directly adjusting predictions based on the overall (or subgroup-specific) incidence difference (on the clog-log scale). This approach, where stratification is optional, provides flexibility and may be particularly advantageous when detailed subgroup data are unavailable or unreliable, or when a straightforward adjustment for baseline risk differences is sufficient.

This approach assumes that the primary difference between the two populations lies in the baseline hazard, the relative effects of risk factors (eg, hazard ratios for smoking or hypertension) are transferable between populations and individual-level cohort data are unavailable to re-estimate covariate coefficients [18]. For CVD risk prediction, the relative effect of risk factors is usually consistent across populations [3], as reflected in our findings where risk factor hazard ratios were similar between UKB and CHERRY. Original SNNs proportionally underpredicted risk across each risk decile in CHERRY, indicating a systemic bias in risk estimation. Our population-level recalibration method effectively addressed the risk overestimation observed in our external validation. This approach is advantageous when individual-level data are scarce but aggregate incidence is known, particularly if miscalibration primarily reflects a shift in baseline risk. However, as other methods focused on intercept adjustment (eg, updating baseline hazard in Cox models), it may not fully rectify complex miscalibration patterns (eg, varying across risk levels or reversing direction), especially when the model’s discrimination is suboptimal. This was demonstrated by our simulation analysis ([Supplementary Methods](#)). Besides, it is important to recognize that models developed using observational data from populations where risk-based treatments are prevalent likely predict risk conditional on existing treatment patterns, rather than the “treatment-free” risk. Differences in such patterns between development and

validation populations can also contribute to model miscalibration and highlight the need for recalibration approaches like the one evaluated in this paper. It's worth noting that we calculated the rescaling factors for population-level recalibration method using the entire validation set in this study, which is both appropriate and desirable because it maximizes the stability of the incidence estimate when the sample size is large and representative of the target population ([Supplementary Methods](#)). However, this may not always be the case; in particular, using the full validation set for recalibration in smaller or less representative cohorts can increase the risk of overfitting. In such situations, it is advisable to validate the recalibrated model on a separate hold-out sample.

Comparing SNN models' performance against a simple Cox model, conceptually similar to widely used tools like the FRS, provides a practical context for assessing the clinical implications of recalibrating SNNs in new populations. The absolute increment in the C-index for recalibrated DeepSurv compared to the simple Cox model was 0.0042 in the CHERRY. The improved discrimination is due to its ability to capture nonlinear interactions among predictors. The seemingly modest increase in the C-index for DeepSurv is notable when compared to gains from adding biomarkers to Cox models, such as high-sensitivity troponin I, N-terminal pro-B-type natriuretic peptide, and C-reactive protein, which resulted in increases of 0.0039, 0.0049, and 0.0031, respectively, for the 10-year CVD risk prediction [33]. Moreover, DeepSurv could correctly reclassify 666 of 9629 CVD cases into the high-risk group. Even small improvements in discrimination can refine risk stratification in clinical settings.

Our findings regarding the modest improvement in SNNs over traditional Cox models differ from previous research due to variations in predictors and models. A recent meta-analysis found ML algorithms offer only a slight advantage over traditional CVD scores, with a small increase in pooled C-indices [34]. For instance, Yu et al. [14] found minor improvements (C-indices increase of 0.003) using SNNs with 8 predictors compared to Cox in White male subgroups. But other studies, like Dolezalova et al. [12] reported no significant difference (C index: 0.7446 vs 0.7443) in CVD risk prediction with 47 predictors. These variations suggest that, even with more predictors, the simpler correlations in tabular data may limit the benefits of SNNs compared to traditional models, unlike the more complex relationships seen in images, genetic information [11], diagnostic text data [10], or repeated measurements [35].

Risk estimation for the same patient can vary significantly across different models, as shown in a simulation study comparing international CVD risk scores [36]. In our study, the age-specific DeepSurv and Cox model showed a 60%-80% overlap in high-risk subgroups and they identified high-risk patients with distinct characteristics, demonstrating how varying predictor weights can influence risk identification. Such variability may confuse treatment recommendations for clinicians and patients.

Ensemble learning could help resolve this by integrating complementary information from different models [37,38]. With the growing use of EHRs, integrating multiple ML models into health systems would enable easy updates based on changing risks [39], as per our proposed recalibrated method. This could enhance personalized decision-making and health care quality by delivering timely information to patients.

We acknowledge several limitations. First, although SNNs can accommodate missing data via preprocessing techniques like imputation, our study used a complete-case analysis, which may introduce selection bias. This subset might represent patients with more complete data, potentially those already engaged in risk management. Further work to investigate the influence of different imputation strategies on SNN models' performance and recalibration is still needed. Second, we used fully connected neural networks, which is appropriate for comparison with statistical models with limited predictors. We also did not address competing mortality risks in modeling due to their low rate in our study. All of these simplified the primary focus on evaluating the SNNs' recalibration method. Future studies could explore more complex scenarios, such as incorporating deep learning approaches with multimodal data to enhance performance and applying algorithms like DeepHit or multitask learning to handle competing risks. Furthermore, our population-level recalibration method assumes that the relative difference in baseline risk between the development and target populations is constant over the calendar period represented in the validation data. Temporal changes in the risk factor prevalence or health care could violate this assumption. Future research could explore dynamic recalibration approaches or methods incorporating time-varying adjustment factors, particularly for prospective model implementation and monitoring. Lastly, a limitation of our study is that the statistical analysis plan for this study was not prospectively registered; however, the study protocol for the CHERRY cohort was previously published [20].

5. Conclusions

This study highlights the critical need for recalibrating SNN models to specific target populations. The proposed recalibration method, which uses population-level summarized data, significantly improves the accuracy of model predictions. This approach provides a new perspective to implementing ML in survival prediction and disease prevention, enabling more effective applications across diverse demographic settings.

CRedit authorship contribution statement

Xiaofei Liu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data

curation, Conceptualization. **Jun Zhang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Larry Liu:** Writing – review & editing, Supervision. **Xun Tang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Pei Gao:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

P.G. reported receiving research funds from Bayer, but the funding sources had no relation to this study. There are no competing interests for any other author.

Acknowledgments

We thank the Health and Family Planning Bureau of Yinzhou District for providing access to the administrative databases used in the study and all the CHERRY study investigators for their contributions.

Supplementary data

Supplementary data related to this article can be found at doi:[10.1016/j.jclinepi.2025.111895](https://doi.org/10.1016/j.jclinepi.2025.111895).

Data availability

The authors do not have permission to share data.

References

- [1] Arnett DK, Blumenthal RS, Albert MA, Buroker AB, Goldberger ZD, Hahn EJ, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American college of cardiology/American heart association task force on clinical practice guidelines. *Circulation* 2019;140:e596–646.
- [2] D'Agostino RB, Vasan RS, Pencina MJ, Wolf PA, Cobain M, Massaro JM, Kannel WB. General cardiovascular risk profile for use in primary care: the Framingham Heart Study. *Circulation* 2008;117:743–53.
- [3] WHO CVD Risk Chart Working Group. World Health Organization cardiovascular disease risk charts: revised models to estimate risk in 21 global regions. *Lancet Glob Health* 2019;7:e1332–45.
- [4] Goldstein BA, Navar AM, Carter RE. Moving beyond regression techniques in cardiovascular risk prediction: applying machine learning to address analytic challenges. *Eur Heart J* 2017;38:1805–14.
- [5] Katzman JL, Shaham U, Cloninger A, Bates J, Jiang T, Kluger Y. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Med Res Methodol* 2018;18:24.
- [6] Lee C, Zame W, Yoon J, van der Schaar M. DeepHit: a deep learning approach to survival analysis with competing risks. *Proc AAAI Conf Artif Intelligence* 2018;32. <https://doi.org/10.1609/aaai.v32i1.11842>.
- [7] Gensheimer MF, Narasimhan B. A scalable discrete-time survival model for neural networks. *PeerJ* 2019;7:e6257.
- [8] Nagpal C, Li X, Dubrawski A. Deep survival machines: fully parametric survival regression and representation learning for censored data with competing risks. *IEEE J Biomed Health Inform* 2021;25:3163–75.
- [9] Cai Y, Cai Y-Q, Tang L-Y, Wang YH, Gong M, Jing TC, et al. Artificial intelligence in the risk prediction models of cardiovascular disease and development of an independent validation screening tool: a systematic review. *BMC Med* 2024;22:56.
- [10] Barbieri S, Mehta S, Wu B, Bharat C, Poppe K, Jorm L, Jackson R. Predicting cardiovascular risk from national administrative databases using a combined survival analysis and deep learning approach. *Int J Epidemiol* 2022;51:931–44.
- [11] Steinfeldt J, Buergel T, Look L, Kittner P, Ruyoga G, Zu Belzen JU, et al. Neural network-based integration of polygenic and clinical information: development and validation of a prediction model for 10-year risk of major adverse cardiac events in the UK Biobank cohort. *Lancet Digit Health* 2022;4:e84–94.
- [12] Dolezalova N, Reed AB, Despotovic A, Obika BD, Morelli D, Aral M, Plans D. Development of an accessible 10-year Digital CardioVascular (DiCAVA) risk assessment: a UK Biobank study. *Eur Heart J Digit Health* 2021;2:528–38.
- [13] Hathaway QA, Yanamala N, Budoff MJ, Sengupta PP, Zeb I. Deep neural survival networks for cardiovascular risk prediction: the Multi-Ethnic Study of Atherosclerosis (MESA). *Comput Biol Med* 2021;139:104983.
- [14] Deng Y, Liu L, Jiang H, Peng Y, Wei Y, Zhou Z, et al. Comparison of state-of-the-art neural network survival models with the pooled cohort equations for cardiovascular disease risk prediction. *BMC Med Res Methodol* 2023;23:22.
- [15] Kavousi M, Leening MJG, Nanchen D, Greenland P, Graham IM, Steyerberg EW, et al. Comparison of application of the ACC/AHA guidelines, Adult Treatment Panel III guidelines, and European Society of Cardiology guidelines for cardiovascular disease prevention in a European cohort. *JAMA* 2014;311:1416–23.
- [16] Yang X, Li J, Hu D, Chen J, Li Y, Huang J, et al. Predicting the 10-year risks of atherosclerotic cardiovascular disease in Chinese population: the China-par project (prediction for ASCVD risk in China). *Circulation* 2016;134:1430–40.
- [17] Van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW, Topic Group 'Evaluating diagnostic tests and prediction models' of the STRATOS initiative. Calibration: the Achilles heel of predictive analytics. *BMC Med* 2019;17:230.
- [18] Pennells L, Kaptoge S, Wood A, Sweeting M, Zhao X, White I, et al. Equalization of four cardiovascular risk algorithms after systematic recalibration: individual-participant meta-analysis of 86 prospective studies. *Eur Heart J* 2019;40:621–31.
- [19] Sudlow C, Gallacher J, Allen N, Beral V, Burton P, Danesh J, et al. UK biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. *PLoS Med* 2015;12:e1001779.
- [20] Lin H, Tang X, Shen P, Zhang D, Wu J, Zhang J, et al. Using big data to improve cardiovascular care and outcomes in China: a protocol for the Chinese Electronic health Records Research in Yinzhou (CHERRY) Study. *BMJ Open* 2018;8:e019698.
- [21] Hippisley-Cox J, Coupland C, Brindle P. Development and validation of QRISK3 risk prediction algorithms to estimate future risk of cardiovascular disease: prospective cohort study. *BMJ* 2017;357:j2099.
- [22] Riley RD, Collins GS, Ensor J, Archer L, Booth S, Mozes SI, et al. Minimum sample size calculations for external validation of a clinical prediction model with a time-to-event outcome. *Stat Med* 2022;41:1280–95.
- [23] Riley RD, Archer L, Snell KIE, Ensor J, Dhiman P, Martin GP, et al. Evaluation of clinical prediction models (part 2): how to undertake an external validation study. *BMJ* 2024;384:e074820.

- [24] Pencina MJ, D'Agostino RB, D'Agostino RB, Vasan RS. Evaluating the added predictive ability of a new marker: from area under the ROC curve to reclassification and beyond. *Stat Med* 2008;27:157–72.
- [25] Chambless LE, Cummiskey CP, Cui G. Several methods to assess improvement in risk prediction models: extension to survival analysis. *Stat Med* 2011;30:22–38.
- [26] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378.
- [27] van Smeden M, Heinze G, Van Calster B, Asselbergs FW, Vardas PE, Bruining N, et al. Critical appraisal of artificial intelligence-based prediction models for cardiovascular disease. *Eur Heart J* 2022;43:2921–30.
- [28] Li Y, Sperrin M, Ashcroft DM, van Staa TP. Consistency of variety of machine learning and statistical models in predicting clinical risks of individual patients: longitudinal cohort study using cardiovascular disease as exemplar. *BMJ* 2020;371:m3919.
- [29] Ishwaran H, Kogalur UB, Chen X, Minn AJ. Random survival forests for high-dimensional data. *Stat Anal Data Mining: An ASA Data Sci J* 2011;4:115–32.
- [30] Kaur I, Doja MN, Ahmad T. Data mining and machine learning in cancer survival research: an overview and future recommendations. *J Biomed Inform* 2022;128:104026.
- [31] Binuya MaE, Engelhardt EG, Schats W, Schmidt MK, Steyerberg EW. Methodological guidance for the evaluation and updating of clinical prediction models: a systematic review. *BMC Med Res Methodol* 2022;22:316.
- [32] Meijerink LM, Dunias ZS, Leeuwenberg AM, de Hond AAH, Jenkins DA, Martin GP, et al. Updating methods for artificial intelligence-based clinical prediction models: a scoping review. *J Clin Epidemiol* 2025;178:111636.
- [33] Neumann JT, Twerenbold R, Weimann J, Ballantyne CM, Benjamin EJ, Costanzo S, et al. Prognostic value of cardiovascular biomarkers in the population. *JAMA* 2024;331:1898–909.
- [34] Liu W, Laranjo L, Klimis H, Chiang J, Yue J, Marschner S, et al. Machine-learning versus traditional approaches for atherosclerotic cardiovascular risk prognostication in primary prevention cohorts: a systematic review and meta-analysis. *Eur Heart J Qual Care Clin Outcomes* 2023;9:310–22.
- [35] Li C, Liu X, Shen P, Sun Y, Zhou T, Chen W, et al. Improving cardiovascular risk prediction through machine learning modelling of irregularly repeated electronic health records. *Eur Heart J Digit Health* 2024;5:30–40.
- [36] Mancini GBJ, Ryomoto A, Yeoh E, Brunham LR, Hegele RA. Recommendations for statin management in primary prevention: disparities among international risk scores. *Eur Heart J* 2024;45:117–28.
- [37] Lee C, Light A, Alaa A, Thurtle D, van der Schaar M, Gnanapragasam VJ. Application of a novel machine learning framework for predicting non-metastatic prostate cancer-specific mortality in men using the Surveillance, Epidemiology, and End Results (SEER) database. *Lancet Digit Health* 2021;3:e158–65.
- [38] Alaa AM, Bolton T, Di Angelantonio E, Rudd JHF, van der Schaar M. Cardiovascular disease risk prediction using automated machine learning: a prospective study of 423,604 UK Biobank participants. *PLoS One* 2019;14:e0213653.
- [39] Liu X, Li Q, Chen W, Shen P, Sun Y, Chen Q, et al. A dynamic risk-based early warning monitoring system for population-based management of cardiovascular disease. *Fundam Res* 2021;1:534–42.